Risk factors for human immunodeficiency virus among blood donors in Cameroon: evidence for the design of an Africa-specific donor history questionnaire

Claude T. Tagny,[1,2] Georges Nguefack-Tsague,[3] Diderot Fopa,[2] Celestin Ashu,[2] Estel Tante,[4] Pauline Ngo Balogog,[5] Olivier Donfack,[6] Dora Mbanya,[1,2] Syria Laperche,[7] and Edward Murphy[8]

BACKGROUND: In sub-Saharan Africa improving the deferral of at-risk blood donors would be a cost-effective approach to reducing transfusion-transmitted human immunodeficiency virus (HIV) infections. We performed a pilot case-control study to identify the risk factors for HIV infection and to develop an adapted donor history questionnaire (DHQ) for sub-Saharan Africa.

STUDY DESIGN AND METHODS: We recruited 137 HIV-positive donors (cases) and 256 HIV-negative donors (controls) and gathered risk factor data using audio computer-assisted self-interview. Variables with univariate associations were entered into a logistic regression model to assess independent associations. A scoring scheme to distinguish between HIV-positive and HIV-negative donors was developed using receiver operating characteristics curves.

RESULTS: We identified 16 risk factors including sex with sex worker, past history or treatment for sexually transmitted infections, and having a partner who used injected or noninjected illegal drugs. Two novel risks were related to local behavior: polygamy (odds ratio [OR], 22.7; 95% confidence interval [CI], 5.9-86.7) and medical or grooming treatment on the street (OR, 1.8; 95% CI, 1.0-3.0). Using the 16 selected items the mean scores (>100) were 82.6 ± 6.7 (range, 53.2-95.1) and 85.1 ± 5.2 for HIV-negative donors versus 77.9 ± 6.8 for HIV-positive ones ($p = 0.000$). Donors who scored between 80 and 90 were more likely to be HIV negative than those who scored less (OR, 31.4; 95% CI, 3.1-313.9).

CONCLUSION: We identified both typical and novel HIV risk factors among Cameroonian blood donors. An adapted DHQ and score that discriminate HIV-negative donors may be an inexpensive means of reducing transfusion-transmitted HIV through predonation screening.

To increase the safety of blood products transfused in Africa, three strategies need to be considered: more effective medical selection of blood donors, better laboratory detection of transfusion-transmitted infections (TTIs), and physical and chemical treatment of blood products. Detection of TTIs has greatly improved in most African settings during the past decade but residual risk is still high.[1] The efficacy of serologic screening tests is still limited due to many reasons such as viral diversity, inappropriate or incomplete quality assurance in many centers, and inability to detect

ABBREVIATIONS: DHQ(s) = donor history questionnaire(s); INTS = Institut National de la Transfusion Sanguine; MSM = men who have sex with men; ROC = receiver operating characteristics; STD = sexually transmitted disease; TTI(s) = transfusion-transmitted infection(s).

From the [1]Department of Hematology and the [3]Department of Public Health, Faculty of Medicine and Biomedical Sciences of University of Yaoundé I; and the [2]Hematology & Transfusion Service University Teaching Hospital, Yaoundé, Cameroon; the [4]Bamenda Regional Hospital, Bamenda, Cameroon; [5]Essos Hospital Center, Yaoundé, Cameroon; the [6]Department of Public Health, Institute of Tropical Medicine, Antwerp, Belgium; the [7]Institut National de la Transfusion Sanguine, Paris, France; and [8]Blood Systems Research Institute, San Francisco, California.

Address reprint requests to: Claude Tayou Tagny, MD, MS, MSc, Department of Haematology, Faculty of Medicine and Biomedical Sciences, University of Yaoundé, PO Box 5739, Yaoundé, Cameroon; e-mail: tayouclaude@yahoo.fr.

This study was supported by an unconditional grant from the NBF Scientific Research Grants Program.

Received for publication October 29, 2016; revision received March 5, 2017; and accepted March 18, 2017.

doi:10.1111/trf.14140

© 2017 AABB

TRANSFUSION 2017;57;1912–1921

recently infected subjects. Furthermore, physical-chemical treatment of blood products is still only feasible for platelet concentrate and plasma on a daily basis and most of Francophone African countries cannot currently afford the cost of this type of blood treatment. Thus, improving the medical selection process of blood donors to exclude subjects at high risk of transmitting blood-borne infectious pathogens seems to be an important approach in resources-limited countries.

The current donor history questionnaires (DHQs) used in the Francophone African countries are usually based on risk factors identified elsewhere.[2] That seems inappropriate in such different sociocultural settings. In sub-Saharan Africa, blood donor characteristics have been previously described.[3,4] However, only a few TTI risk factors have been reported in local studies in the past two decades. These include sex with prostitutes in the past 5 years without regular condom use, prostitute contact in the past 5 years regardless of condom use, age of 30 to 39 years, sexually transmitted disease (STD) history, and living in a semiurban or urban area.[5-9] These observations cannot be conclusive because in most of these studies, risk factors had not been identified with a confirmatory screening test. Furthermore, the studies did not consider the possible bias generated by false responses since in-person interviews of blood donors may lead to biased responses, particularly in the context of family blood donation.[10]

One of the most important concerns in blood safety is HIV infection. In Cameroon, the prevalence of HIV is high in blood donors, namely, 1.2% in 2015 at the Yaounde University Teaching Hospital Blood Bank.[11] In our literature review, we could not identify well-performed studies of HIV risk factors in Francophone Africa. In Cameroon, preliminary studies have identified some HIV risk factors in donors.[12,13] HIV seemed to be associated with less frequent blood donation, knowledge of HIV status, and those with occupational exposure to HIV.

It has been shown that HIV risk factors found in blood donors may differ from those found in the general population. Indeed, type of donation (family replacement or remunerated vs. nonremunerated), frequency of donation (first-time vs. regular),[14] and recruitment venue[6] are associated factors specifically related to blood donation activity. Thus, studies in blood donors intending to identify specific risk factors associated with HIV infection have two potential roles: first, to identify risk factors specific to blood donors which can be used in predonation screening questionnaires; and second, recognizing that blood donors are not representative of the general population, data from the study may also reveal hitherto unrecognized HIV risk factors that can be further studied by public health authorities.

This study intended to determine the risk factors associated with HIV infection in Cameroonian blood donors and, by extension, sub-Saharan African blood donors. The data will be used to design a DHQ for an efficient predonation screening in that setting.

We hypothesized that local risk factors for HIV among Cameroonian blood donors include polygamy treatment on the street, scarification, traditional healing, and remunerated blood donation. We also hypothesized that an adapted DHQ developed from selected risk factors will be able to discriminate HIV-positive blood donors with a high sensitivity and specificity.

MATERIALS AND METHODS

This case-control study was conducted from December 2014 to August 2016 in three blood banks in two main sociocultural areas of the country to include diverse behaviors and ethnicities: the blood banks of the University Teaching Hospital of Yaoundé (Site I); the Hospital Center of Essos, Yaoundé (Site II); and that of the North Western Regional Hospital, Bamenda (Site III). HIV-confirmed-seropositive donors (cases) as well as seronegative donors (controls) were enrolled. Cases were recruited by consecutive sampling of all HIV positive cases. Controls were matched in a 2:1 ratio to cases on age, sex, and type of donation. To give the same chance to all consenting and eligible donors to be recruited, the first control who matched with a recruited case was invited to participate.

Inclusion of donors

Subjects were enrolled from December 2014 to March 2016. Before enrollment, all eligible donors (aged between 18 and 65 years, in good general health, with hemoglobin levels of at least 11 g/dL) had to consent by signing the written informed consent document relative to the study. Autologous blood donors were excluded.

Any blood donor who tested HIV positive according to the screening algorithm of the Yaoundé University Teaching Hospital, which combined a rapid HIV antibody assay (Determine HIV-1/-2, Inverness Medical France) and an antigen-antibody combination assay (Murex HIV Ag/Ab, DiaSorin)[15] was called back for counseling and notification. Blood donors were informed about their serology and, if they consented, were enrolled in the study. For each case, two HIV-negative (on the basis of the same testing algorithm), age- and sex-matched donors (controls) were systematically called back. These were informed about their negative HIV status, and if they consented, were also enrolled into the study. Twice the number were called back to make up for any refusals to participate, uncompleted questionnaires, low quality of some samples, and false positivity of results. Based on a sample size calculation, we planned a target enrollment of 115 cases and 230 controls. The study protocol was reviewed and

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

approved by the Yaoundé University Teaching Hospital and the Cameroonian National Ethical Committee.

Questionnaire design

We selected and assessed reported HIV risk factors in sub-Saharan African blood donors from previous studies[13,16-18] and used a more rigorous approach. The risks identified in this study were used to design a questionnaire and to test its performance. The scoring scheme was used to test the performance of the questionnaire against results of the standard laboratory method for HIV screening in distinguishing HIV-positive donors from HIV-negative donors. The best threshold (measured by the total score) was selected by receiver operating characteristics (ROC) curves based on calculated sensitivity, specificity, and correct classification rate. Computer software (IBM-SPSS 21, IBM Corp.; and Stata, Version 14, StataCorp) were used for statistical analysis.

A detailed HIV risk factor questionnaire of 30 questions was administered to all enrolled subjects. Since a self-administered computerized questionnaire may increase risk reporting by blood donors, we chose this procedure for donor questionnaire assessment.[19] A self-administered bilingual (English and French) audio computer-assisted self-interview on a laptop computer was used to maximize reporting of stigmatized behaviors. Before this study, we evaluated a paper version of the 15- to 20-minute questionnaire on 25 blood donors and concluded that the questions were comprehensible without much need of explanation. Each questionnaire was administered only at the blood bank, in an isolated area appropriate for confidentiality.

Definitions included the following: "living together," the fact that two sexual partners share the same house for a long time but are not officially married. Having a "treatment on the street" is having undergone medical, dental (e.g., dental extraction), or cosmetic (e.g. manicure, pedicure) procedure with a surgical instrument or bloody material in an informal setting such as street, that is, by an unlicensed practitioner. "Polygamy" has been considered as a man living together with two female sexual partners or more in an official or an unofficial marital status. "Excision/female genital cutting" is removal of the clitoris often traditionally conducted using nonsterile implements.

Laboratory screening

At the time of the enrollment and interview, 15 and 10 mL of blood were collected from blood donors found HIV positive and HIV negative at the screening stage, respectively. All samples were locally stored at −20°C until shipped in the appropriate transport conditions to the National Reference Center (Institut National de la Transfusion Sanguine [INTS]) to be tested with enzyme immunoassay antigen-antibody HIV-1 and -2 assay (Genscreen Ag/Ab Ultra BioRad) and with supplemental assays (WB HIV Blot 2.2, Abbott; Geenius HIV, BioRad) and/or HIV RNA polymerase chain reaction (Cobas TaqMan HIV-1, Roche Molecular Diagnostics; limit of detection, 47 copies/mL) to avoid erroneous classification, according to the algorithm detailed in Fig. 1. Samples were considered as HIV positive if they were reactive on WB and/or RNA and negative when both tests provided negative results. Donors who asked for their results were first counseled and notified at the level of Yaoundé ("results to be confirmed") and were called back for a final notification when results from INTS were available.

Statistical analysis

Only completed questionnaires were retained for data analysis. We used a continuity-corrected chi-square statistic or Fisher's exact test to evaluate this null hypothesis.

The dependent variable was HIV status scored as 1 for HIV+ and 0 for HIV–, and items (questions) on the questionnaire were independent variables (potential risk factors) as 4 for "never," 3 for "rarely," 2 for "occasionally," and 1 for "frequently." Since all continuous variables do not deviate significantly from normal distribution, t test was used for comparing means between cases and controls. For univariate analysis, associations of risk factors with HIV status was assessed, using contingency tables, chi-square test, or Fisher's exact test when appropriate. The strength of an association was assessed using adjusted odds ratios (ORs) with 95% confidence interval (CI) using conditional (fixed effects) multivariate logistic regression after controlling (adjusting) for level of education and marital status. Factors with p values less than 0.05 were considered potential risk factors and were qualified for a scoring scheme. The internal consistency (reliability) of the questionnaire was tested by calculating Cronbach's alpha to check if the content is homogeneous, that is, if the selected items were measuring risk factors.

RESULTS

The mean time to interview after donation for cases and controls was 8.6 and 7.9 months for cases and controls, respectively. In total, 471 participants agreed to be recruited and came back for notification. Seventy-three participants were excluded because they stopped the self-administered questionnaire before the end of the procedure. After exclusion of incomplete files (n = 73) and inconclusive confirmatory results (n = 5), 393 (348 were enrolled in Site I, 34 in Site II, and 11 in Site III) were included for the statistical analysis (Fig. 2). These participants originated from all the 10 regions of Cameroon and were distributed in 137 (34.9%) cases and 256 (65.1%) controls. The numbers of HIV-positive/HIV-negative participants per site were 125/223, 10/24, and 2/9 for Sites I, II, and III, respectively. Demographic data are reported in

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

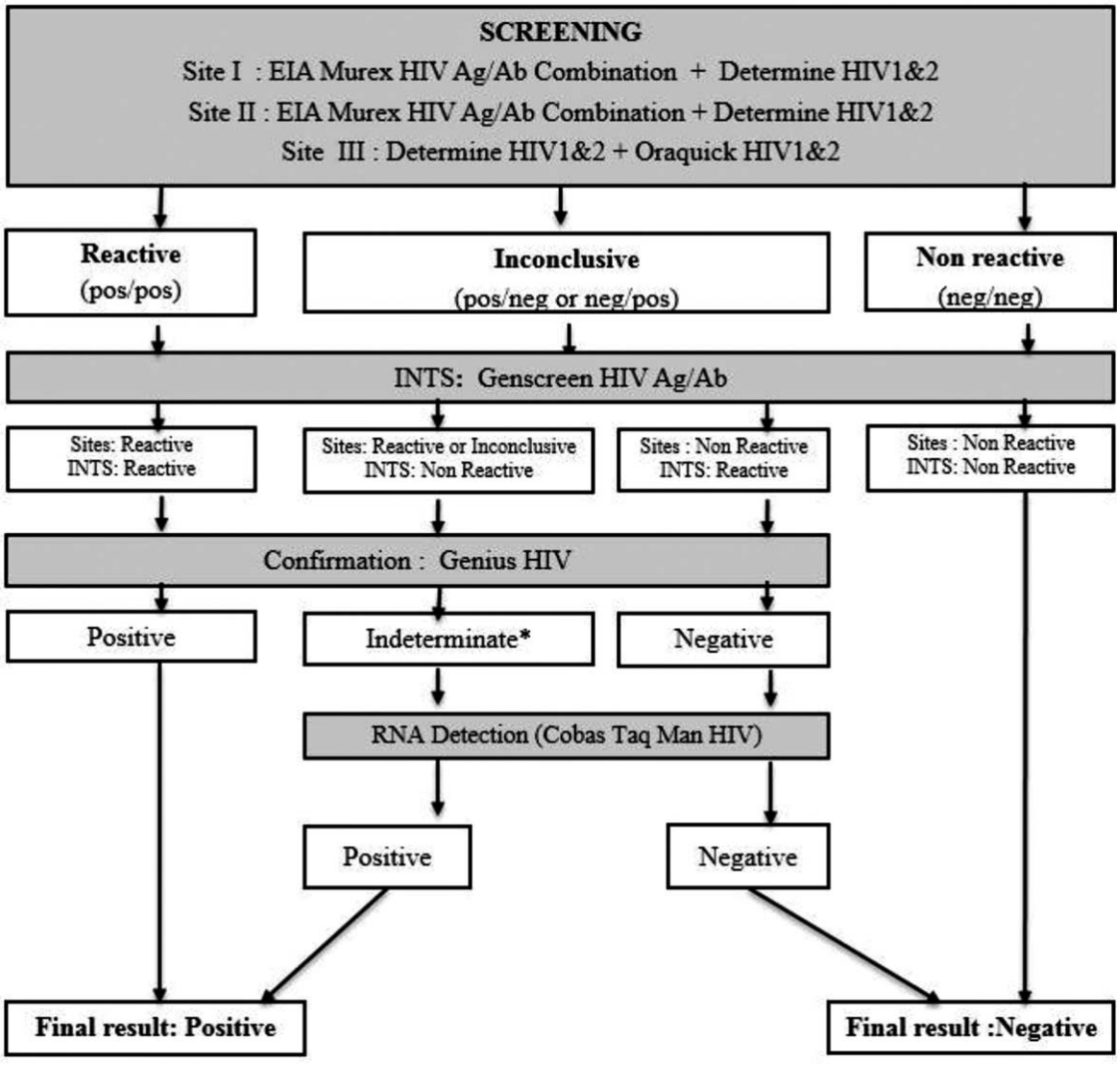


Fig. 1. Algorithm used for HIV diagnosis. *Immunoblot indeterminate and RNA detection not done: samples were classified as inconclusive. EIA = enzyme immunoassay; neg = negative; pos = positive; YUTH = Yaoundé University Teaching Hospital.

Table 1. The mean ± SD age of the donors was 28.9 ± 6.7 (range, 19-59) years: 30.4 ± 6.8 for cases and 29.7 ± 6.7 for controls, with no significant difference ($p = 0.95$). There were 309 (78.6%) males and 84 (21.4%) females, with no significant difference between cases and controls ($p = 0.22$).

Risk factors

We identified 16 risk factors including sex with sex worker, having been treated for sexually transmitted infections, ever having had a STD, and having a partner who ever used injected or not injected illegal drugs. Two more were related to local behavior: having more than one wife (polygamy) was strongly associated with HIV infection with OR of 22.7 (95% CI, 5.9-86.7) for two wives, and treatment on the street such as pedicure and/or manicure or tooth care ($p = 0.0095$) with a somewhat lower OR (1.8;

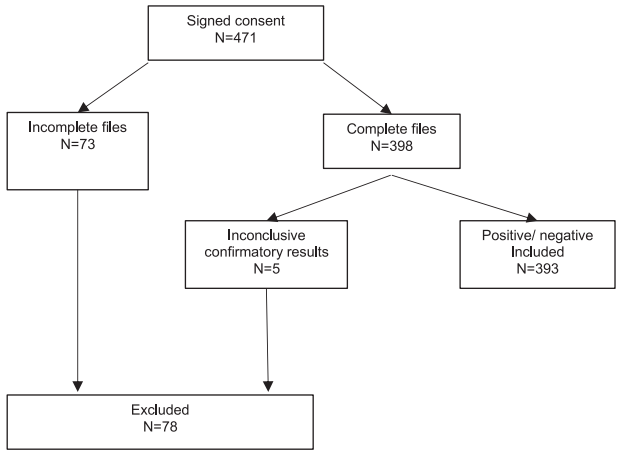


Fig. 2. Recruitment and selection of participants.

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Demographics characteristics

Characteristic	HIV negative (n = 256)*	HIV positive (n = 137)*	Total (n = 393)*	OR (95% CI)	p value
Sex (p = 0.22)					
Female	50 (19.5)	34 (24.8)	84 (21.4)	1	
Male	206 (80.5)	103 (75.2)	309 (78.6)	1.0 (0.18-5.57)	1.00
Age (years; p = 0.95)					
18-20	4 (1.6)	2 (1.5)	6 (1.5)	1	
21-30	162 (63.3)	81 (59.1)	243 (61.8)	1.0 (0.18-5.57)	1.0
31-40	71 (27.7)	43 (31.4)	114 (29.0)	1.21 (0.21-6.89)	0.83
41-50	14 (5.5)	8 (5.8)	22 (5.6)	1.14 (0.17-7.69)	0.89
51-60	5 (2.0)	3 (2.2)	8 (2.0)	1.2 (0.13-11.05)	0.87
Level of education (p = 0.0001)					
None	3 (1.2)	1 (0.7)	4 (1.0)	1	
Primary	20 (8.1)	11 (8.1)	31 (8.1)	3.15 (0.26-38.36)	0.37
Secondary	84 (34.1)	75 (55.1)	159 (41.6)	5.84 (0.53-63.98)	0.15
University	139 (56.5)	49 (36.0)	188 (49.2)	1.83 (0.18-19.12)	0.61
Region of origin (p = 0.000)					
Far North	18 (7.0)	5 (3.6)	23 (5.9)	1	
North	2 (0.8)	2 (1.5)	4 (1.0)	5.78 (0.56-59.28)	0.14
Adamaoua	2 (0.8)	3 (2.2)	5 (1.3)	5.22 (0.65-41.93)	0.12
Center	79 (30.9)	54 (39.4)	133 (33.8)	2.97 (0.96-9.23)	0.059
East	5 (2.0)	4 (2.9)	9 (2.3)	2.56 (0.43-15.28)	0.30
South	11 (4.3)	24 (17.5)	35 (8.9)	8.58 (2.37-31.12)	0.001
Littoral	13 (5.1)	9 (6.6)	22 (5.6)	2.83 (0.75-10.74)	0.126
South West	1 (0.4)	4 (2.9)	5 (1.3)	14.91 (1.19-187.4)	0.036
West	116 (45.3)	28 (20.4)	144 (36.6)	0.91 (0.30-2.78)	0.86
North West	9 (3.5)	4 (2.9)	13 (3.3)	2.00 (0.39-10.28)	0.40
Marital status (p = 0.003)					
Single	178 (69.8)	73 (54.1)	251 (64.4)	1	
Living together	26 (10.2)	33 (24.4)	59 (15.1)	3.16 (1.70-5.86)	0.000
Married	47 (18.4)	27 (20.0)	74 (19.0)	1.54 (0.79-3.0)	0.20
Divorced or widowed	4 (1.6)	2 (1.5)	6 (1.5)	1.30 (0.20-8.36)	0.78
Type of donation (p = 0.16)					
Voluntary nonremunerated	109 (42.9)	68 (50.4)	177 (45.5)	1	
Family or replacement	138 (54.3)	66 (48.9)	204 (52.4)	0.69 (0.44-1.08)	0.11
Paid	8 (2.9)	1 (0.7)	9 (2.2)	0.18 (0.02-1.55)	0.12
Number of donations for the past 12 months (p = 0.13)					
0	3 (2.1)	1 (1.5)	4 (1.9)	1	
1	50 (34.5)	13 (19.4)	63 (29.7)	0.63 (0.05-7.70)	0.72
2	90 (62.1)	51 (76.1)	141 (66.5)	1.82 (0.18-17.92)	0.61
3	2 (1.4)	2 (3.0)	4 (1.9)	2.11 (0.06-77.48)	0.68

*Data are reported as number (%).

95% CI, 1.0-3.0). Multivariate analysis results are reported in Table 2. The 16 potential risk factors are presented in Table 3.

Scoring scheme

Using the nonparametric ROC curve, the area under curve was 0.8 (0.7-0.8) leading to 82.2 as cutoff with 77.3% sensitivity and 73.7% specificity, with correct classification of 76.1%. Using the 16 selected items the mean ± SD score (>100) was 82.6 ± 6.7 (range, 53.2-95.1): 85.1 ± 5.2 for HIV-negative donors vs. 77.9 ± 6.8 for HIV-positive ones (p = 0.000). Thus on average, HIV-negative donors scored 7.1 (5.9-8.3) points more than positive ones (p = 0.000). The Cronbach's alpha was 0.58 showing consistency between the items. The logistic regression showed that for an increase of the score of 1%, donors are 1.2 times more likely to be HIV negative (OR, 1.2; 95% CI, 1.1-1.3; p = 0.000; Fig. 3).

The risk of being HIV negative increased linearly with the score (p = 0.000). The score varied between 53 (corresponding to the least risk of being HIV negative) and 96 (corresponding to the highest risk of being HIV negative). Donors who scored between 80 and 90 were 31 times (OR, 31.4; 95% CI, 3.1-313.9; p = 0.003) more likely to be HIV negative than others having a score of less than 80. Likewise, the risk was 133 (OR, 133.4; 95% CI, 10.8-1637.7; p = 0.000) for those who scored between 90 and 96 (Table 4).

DISCUSSION

Most of risk factors identified were related to sexual contact. We found recognized HIV risk factors including sex with sex worker, having been treated for sexually transmitted infections, ever having had a STD, and having a partner who ever used injected or not injected illegal drugs.

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Multivariate risk factors analysis

Risk factor	HIV negative (n = 256)	HIV positive (n = 137)	Total (%)	OR (95% CI)	aOR (95% CI)
Oral sex during the past 12 months (p = 0.11)					
Never	163 (63.7)	70 (51.1)	223 (59.3)	1	1
Rarely	35 (13.7)	27 (19.7)	62 (15.8)	1.89 (1.02-3.54)	1.94 (0.96-3.93)
Occasionally	39 (15.2)	26 (19.0)	65 (16.5)	1.54 (0.86-2.77)	1.49 (0.80-2.78)
Frequently	19 (7.4)	14 (10.2)	33 (8.4)	1.78 (0.83-3.82)	1.90 (0.78-4.60)
Sexual intercourse without condom during the past 12 months (p = 0.00)					
Never	99 (38.7)	24 (17.5)	123 (31.3)	1	1
Rarely	42 (16.4)	17 (12.4)	59 (15.0)	1.45 (0.71-2.97)	1.90 (0.85-4.25)
Occasionally	45 (17.6)	32 (23.4)	77 (19.6)	2.69 (1.44-5.01)	3.59 (1.74-7.43)
Frequently	70 (27.3)	64 (46.7)	134 (34.1)	3.71 (2.0)	3.78 (1.90-7.54)
Anal sex during the past 12 months (p = 0.59)					
Never	239 (93.4)	125 (91.2)	364 (92.6)	1	1
Rarely	5 (2.0)	6 (4.4)	11 (2.8)	2.37 (0.65-8.66)	2.99 (0.48-18.73)
Occasionally	7 (2.7)	3 (2.2)	10 (2.5)	0.80 (0.21-3.14)	1.19 (0.20-7.30)
Frequently	5 (2.0)	3 (2.2)	8 (2.0)	1.26 (0.30-5.32)	1.48 (0.17-12.99)
Sex with people you are not officially married to during the past 12 months (p = 0.000)					
Never	102 (39.8)	37 (27.0)	139 (35.4)	1	1
Rarely	37 (14.5)	14 (10.2)	51 (13.0)	0.89 (0.43-1.85)	1.01 (0.45-2.27)
Occasionally	51 (19.9)	20 (14.6)	71 (18.1)	1.04 (0.55-1.99)	0.95 (0.46-1.96)
Frequently	66 (25.8)	66 (48.2)	132 (33.6)	2.81 (1.70-4.68)	2.56 (1.46-4.49)
Not using condoms with partners you are not officially married to this past 12 months (p = 0.000)					
Never	86 (33.6)	18 (13.1)	104 (26.5)	1	1
Rarely	30 (11.7)	54 (39.4)	84 (21.4)	8.40 (4.06-17.37)	7.41 (3.36-16.37)
Occasionally	20 (7.8)	21 (15.3)	41 (10.4)	4.86 (2.08-11.34)	3.66 (1.49-8.95)
Frequently	120 (46.9)	44 (32.1)	164 (41.7)	1.88 (0.98-3.59)	1.87 (0.91-3.81)
Sex with sex workers during the past 12 months (p = 0.000)					
Never	228 (89.1)	98 (71.5)	326 (83.0)	1	1
Rarely	15 (5.9)	24 (17.5)	39 (9.9)	4.66 (2.28-9.95)	4.07 (1.80-9.21)
Occasionally	9 (3.5)	12 (8.8)	21 (5.3)	4.67 (1.70-12.84)	6.44 (2.03-20.40)
Frequently	4 (1.6)	3 (2.2)	7 (1.8)	1.71 (0.38-7.82)	2.39 (0.41-13.89)
Sex with drug users during the past 12 months (p = 0.000)					
No	253 (98.8)	119 (86.9)	372 (94.7)	1	1
Yes	3 (1.2)	18 (13.1)	21 (5.3)	11.08 (3.25-37.74)	9.96 (2.64-37.60)
Sex with HIV-positive person (p = 0.000)					
No	251 (98.0)	101 (73.7)	352 (89.6)	1	1
Yes	5 (2.0)	36 (26.3)	41 (10.4)	20.17 (6.19-65.74)	20.22 (5.75-71.08)
Sex with HIV-positive person on medication (p = 0.000)					
No	250 (97.7)	118 (86.1)	368 (93.6)	1	1
Yes	6 (2.3)	19 (13.9)	25 (6.4)	6.29 (2.33-16.98)	5.08 (1.78-14.56)
Sex with MSM (p = 0.335)					
No	256 (100.0)	136 (99.3)	392 (99.7)	NA	NA
Yes	0 (0.0)	1 (0.7)	1 (0.3)	NA	NA
Sex with somebody who has already received transfusion (p = 1.0)					
No	250 (97.7)	134 (97.8)	384 (97.7)	1	1
Yes	6 (2.3)	3 (2.2)	9 (2.3)	1.0 (0.25-4.0)	0.84 (0.17-4.12)
Sex with somebody who spent at least a night in jail (p = 0.03)					
No	253 (98.8)	129 (94.2)	382 (97.2)	1	1
Yes	3 (1.2)	8 (5.8)	11 (2.8)	4.0 (1.04-15.30)	3.86 (0.88-16.85)
Sex with somebody whose work involves exposure to blood (p = 0.95)					
No	235 (91.8)	126 (92.0)	361 (91.9)	1	1
Yes	21 (8.2)	11 (8.0)	32 (8.1)	0.93 (0.43-2.0)	1.04 (0.42-2.53)
Treated for sexually transmitted infection (p = 0.000)					
Never	189 (73.8)	52 (38.0)	241 (61.3)	1	1
Rarely	23 (9.0)	20 (14.6)	43 (10.9)	3.04 (1.51-6.12)	2.41 (1.07-5.45)
Occasionally	36 (14.1)	31 (22.6)	67 (17.0)	2.76 (1.56-4.87)	3.13 (1.63-5.99)
Frequently	8 (3.1)	34 (24.8)	42 (10.7)	14.03 (5.62-35.02)	22.23 (7.12-69.43)
Contact with liquids that come from human body (p = 0.65)					
Never	130 (50.8)	65 (47.7)	195 (49.6)	1	1
Rarely	64 (25.0)	41 (29.9)	105 (26.7)	1.22 (0.75-1.98)	1.35 (0.78-2.33)
Occasionally	29 (11.3)	18 (13.1)	47 (12.0)	1.20 (0.62-2.33)	1.16 (0.55-2.43)
Frequently	33 (12.9)	13 (9.5)	46 (11.7)	0.80 (0.40-1.60)	0.85 (0.39-1.85)
Use of noninjected illegal drugs (p = 0.01)					
No	248 (96.9)	114 (83.2)	362 (92.1)	1	1
Yes	8 (3.1)	23 (16.8)	31 (7.9)	7.99 (3.02-21.19)	8.0 (2.73-23.44)
Use of injected illegal drugs (p = 0.008)					
No	251 (98.0)	127 (92.7)	378 (96.2)	1	1
Yes	5 (2.0)	10 (7.3)	15 (3.8)	4.0 (1.37-11.70)	5.42 (1.55-18.90)

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2: *Continued*

Risk factor	HIV negative (n = 256)	HIV positive (n = 137)	Total (%)	OR (95% CI)	aOR (95% CI)
Shared needles with others (p = 0.01)					
No	221 (86.3)	104 (75.9)	325 (82.7)	1	1
Yes	35 (13.7)	33 (24.1)	68 (17.3)	2.0 (1.15-3.49)	2.58 (1.36-4.88)
Undergone a surgery or tooth care in the past 12 months (p = 0.85)					
No	236 (92.2)	125 (91.2)	361 (91.9)	1	1
Yes	20 (7.8)	12 (8.8)	32 (8.1)	1.07 (0.51-2.23)	1.08 (0.46-2.51)
Had sex with person with hemophilia					
No	256 (100.0)	137 (100.0)	393 (100.0)	NA	NA
Yes	0 (0.0)	0 (0.0)	0 (0.0)	NA	NA
Treatment on the street such as pedicure and/or manicure or tooth care (p = 0.0095)					
No	204 (79.7)	92 (67.2)	296 (75.3)	1	1
Yes	52 (20.3)	45 (32.8)	97 (24.7)	1.87 (1.16-3.02)	1.80 (1.06-3.03)
Undergone acupuncture (p = 0.12)					
No	255 (99.6)	134 (97.8)	389 (99.0)	1	1
Yes	1 (0.4)	3 (2.2)	4 (1.0)	5.16 (0.53-50.41)	2.39 (0.22-25.97)
Tattoos on yourself (p = 0.001)					
No	246 (96.1)	119 (86.9)	365 (92.9)	1	1
Yes	10 (3.9)	18 (13.1)	28 (7.1)	3.87 (1.67-8.99)	3.61 (1.39-9.41)
Piercings on your body (p = 0.23)					
No	249 (97.3)	130 (94.9)	379 (96.4)	1	1
Yes	7 (2.7)	7 (5.1)	14 (3.6)	1.81 (0.59-5.52)	2.65 (0.68-10.31)
Scarifications on your body (p = 0.32)					
No	237 (92.6)	121 (88.3)	358 (91.1)	1	1
Yes	19 (7.4)	16 (11.7)	35 (8.9)	1.4 (0.72-2.76)	1.36 (0.65-2.85)
Undergone endoscopy (p = 0.09)					
No	247 (96.5)	136 (99.3)	383 (97.5)	1	1
Yes	9 (3.5)	1 (0.07)	10 (2.5)	0.20 (0.026-1.61)	0.71 (0.04-12.35)
Do traditional healing (p = 0.69)					
No	2 (0.8)	2 (1.5)	4 (1.0)	1	1
Yes	254 (99.2)	135 (98.5)	389 (99.0)	0.62 (0.05-6.99)	NA
Not circumcised (male only) (p = 0.08)					
No	186 (95.4)	91 (90.1)	277 (93.6)	1	1
Yes	9 (4.6)	10 (9.9)	19 (6.4)	2.37 (0.88-6.40)	2.74 (0.88-8.54)
Excision (female only) (p = 0.01)					
No	49 (100.0)	30 (85.7)	79 (94.0)	NA	NA
Yes	0 (0.0)	5 (14.3)	5 (6.0)	NA	NA
Number of wives (males only) (p = 0.00)					
0	55 (27.0)	8 (8.0)	63 (20.7)	1	1
1	95 (46.6)	31 (31.0)	126 (41.4)	2.62 (1.04-6.60)	2.71 (0.92-7.99)
2	27 (13.2)	43 (43.0)	70 (23.0)	16.33 (5.62-47.48)	22.75 (5.97-86.70)
3	21 (10.3)	17 (17.0)	38 (12.5)	8.40 (2.74-25.74)	9.62 (2.47-37.47)
4 and more	6 (2.9)	1 (1.0)	7 (2.3)	1.82 (0.17-19.90)	2.66 (0.23-30.78)
Have you inherited a wife? (p = 0.51)					
No	190 (91.8)	96 (94.1)	286 (92.6)	1	1
Yes	17 (8.2)	6 (5.9)	23 (7.4)	0.72 (0.26-1.96)	0.88 (0.43-2.07)

*Data are reported as number (column %).
aOR = adjusted odds ratio.

However, we also identified two local behavioral risk factors in these blood donors: polygamy and treatment on the street (tooth care, manicure, and pedicure). We were not able to make definitive conclusions on men who have sex with men (MSM), acupuncture, and excision as very few participants ($n \leq 5$) reported each behavior.

As in another multivariate analysis study in which cases matched with controls for sex and age group, we did not find a correlation of HIV risk with type of donation (voluntary nonremunerated vs. family or replacement).[20] This is opposite to the findings of many other studies,[13,16-18] but none of those considered sex and age group as potential confounding factors. We showed that level of education is associated with HIV infection as reported previously,[21] but we were not able to identify which level of education is involved in that association; level of knowledge on HIV infection, alcohol consumption, and circumcision may be confounding factors.[21,22] Only nine donors (one case and eight controls) reported being paid to donate blood, not enough to make a valuable statistical analysis of paid donation as a HIV risk factor.

Polygamy was associated with HIV infection as hypothesized. This local behavior is not exactly similar to having multiple sexual partners because the polygamy arrangement is stable and could be assumed to be safer than the former. The use of nonsterilized medical instruments is a recognized transmission route of HIV;[23] we identified clitoral excision as a possible risk factor for HIV,

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Suggested model of DHQ for HIV predonation screening in Cameroon

Questions	Responses:marks	Score
1. How many wives official or not do you have?	0:4; 1:4; 2:3; 3:2; 4 or more:1	
During the past 12 months		
2. How often did you tattoo yourself?	Never:4; rarely:3; occasionally:2; frequently:1	
3. How often have you undergone treatment on the street such as pedicure and/or manicure or tooth care?	Never:4; rarely:3; occasionally:2; frequently:1	
4. How often have you shared needles with others?	Never:4; rarely:3; occasionally:2; frequently:1	
5. How often have you been treated for sexually transmitted infection?	Never:4; rarely:3; occasionally:2; frequently:1	
6. How often do you use injected illegal drugs?	Never:4; rarely:3; occasionally:2; frequently:1	
7. How often have you used noninjected illegal drugs?	Never:4; rarely:3; occasionally:2; frequently:1	
8. How often have you had sex with somebody who spent at least a night in jail?	Never:4; rarely:3; occasionally:2; frequently:1	
9. How often have you had sex with an HIV-positive person?	Never:4; rarely:3; occasionally:2; frequently:1	
10. How often have you had sex with an HIV-positive person on medication?	Never:4; rarely:3; occasionally:2; frequently:1	
11. How often have you had sex with drug users?	Never:4; rarely:3; occasionally:2; frequently:1	
12. How often have you had sex with sex workers?	Never:4; rarely:3; occasionally:2; frequently:1	
13. How often have you *not* used condoms with partners you are not officially married to?	Never:4; rarely:3; occasionally:2; frequently:1	
14. How often have you had sex with people you are not officially married to?	Never:4; rarely:3; occasionally:2; frequently:1	
15. How often have you had sexual intercourse without a condom?	Never:4; rarely:3; occasionally:2; frequently:1	
Female only		
16. Are you excised?	No:2; yes:1	

Total marks: min, 16; max, 62. Score formula (female): total marks × (100/62); score formula (male): total marks × (100/60).

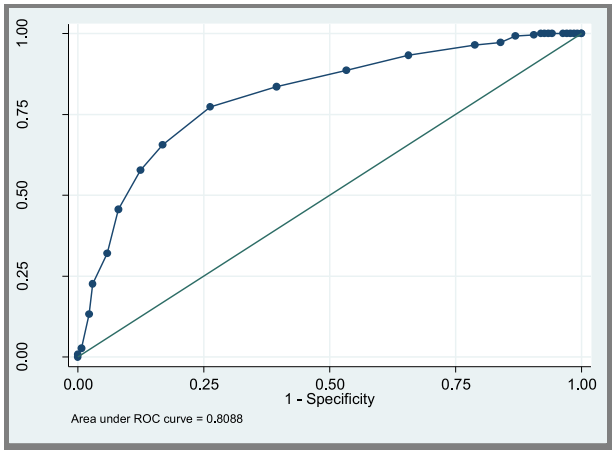


Fig. 3. Nonparametric ROC curve. [Color figure can be viewed at wileyonlinelibrary.com]

most likely by this mechanism. However, the limited number of women (n = 5) who disclosed excision is not enough to make a definitive conclusion. Indeed, traditional healing and scarification were not associated with HIV infection in this study. Traditional healers have been urged to use a new blade for each patient but it is still unclear whether this has limited HIV transmission during scarification or excision.

Sex with a sex worker, having been treated for sexually transmitted infections, ever having had a STD, and having a partner who ever injected drugs were all HIV risk factors that have been reported in studies with similar[20,24,25] or different[13,16,18] methods. We did not find a significant association between exposure to biologic fluids during surgery, tooth care, acupuncture, piercing, or endoscopy. These are all clinic- or hospital-based services where the use of sterilized equipment is expected, so these results are reassuring. Only one participant of 393 (0.2%) reported MSM compared to 133 of 1132 (11.7%) in a similar study in Brazil.[25] Either the frequency of MSM may be lower in Cameroon than in Brazil or in developed countries[24,26,27] or the social stigma in Cameroon may have prevented participants from disclosing this behavior. Moreover, MSM being illegal in Cameroon, some donors may have been afraid to disclose this behavior. Again, the limited number of participants who underwent acupuncture (n = 4) or who admitted MSM is not enough to make a final conclusion. Similarly, only 15 participants (3.8%) reported injection drug use, which is very low compared to studies in Brazil or developed countries.[24,26,27]

The sensitivity and specificity of any DHQ are important because they describe the ability of the questionnaire to identify HIV-positive donors while allowing HIV-negative donors to donate blood. The low sensitivity of DHQ increases the risk of collecting blood from at-risk donors while low specificity would result in the exclusion of too many safe donors and adversely impact the blood supply. Based on our results, a potential donor who scored 80/100 or more on our DHQ would have a high probability of being HIV negative. Based on our calculated specificity at this cutoff, 16.4% of HIV-negative donors would be falsely excluded from donation. We recommend setting the cutoff at a score of 80/100 from which 70% of donors scored and there is the maximum chance to select an HIV-negative donor. Our literature review revealed few assessments of DHQ in sub-Saharan Africa but the few that we did find reported that DHQs are inefficient.[2,28] Our results are therefore an important contribution that needs to be followed up with more work in defining an

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Scoring scheme*

Score	HIV negative	HIV positive	Total	OR (95% CI)	p value
[53,70]	1 (0.4)	13 (9.5)	14 (3.6)	1	
[70,80]	41 (16.0)	70 (51.1)	111 (28.2)	5.76 (0.59-56.17)	0.13
[80,90]	156 (60.9)	50 (36.5)	206 (52.4)	31.41 (3.14-313.91)	0.003
[90,96]	58 (22.7)	4 (2.9)	62 (15.8)	133.40 (10.87-1637.72)	0.000
Total	256 (100.0)	137 (100.0)	393 (100.0)		

*Data are reported as number (%).

accurate DHQ since inadequate blood supply is a critical issue in resource-limited African countries.

Our study used a more rigorous approach compared to others. Diagnosis of HIV infection was based on confirmed testing in a quality-assured laboratory and the effects of confounding were reduced using multivariable analysis that included controlling for residual confounding by the frequency matching variables of sex and age. All the 10 regions of Cameroon are well represented as far as origin of participants are concerned despite lower than projected enrollment at Sites II and III. The use of a computer-based system like audio computer-assisted self-interview created a confidential environment for participants and increased disclosure of HIV risk factors among blood donors.[19,29] Other studies have noted the benefit of a computer-based interview for increasing the efficiency of the DHQ.[30,31]

Limitations of the study include potential response bias in that donors may not have reported all high-risk behavior. Motivations for nondisclosure are difficult to overcome given that donors appeared to understand the purpose of the DHQ, but stigma may still be a factor. Some questions may have not be well understood by participants; for example, sharing of needles was reported more frequently than the use of injected drugs. Participants may have understood needle sharing in a medical instead of drug use context. Clearer formulation may certainly improve the quality of DHQ.

This study has public health and governmental policy implications, not only regarding blood safety, but also as an indicator of current HIV risk factors in the general population, as we brought to light some unanticipated risks, such as polygamy, treatment on the street, and possibly excision. Addressing these issues is more likely to reduce transfusion threats especially when they are included in an appropriate DHQ, but our findings need to be placed within the context that the risk of transmission of infectious agents from allogeneic blood transfusion is probably high in our setting. To validate findings of this pilot assessment, a large multicenter study could enroll participants from several African countries into a similar study. Each participating African blood service would need to create an environment of trust and transparency to increase donors' compliance and improve accuracy of such questionnaire.

ACKNOWLEDGMENTS

We thank Roche for the provision of HIV-RNA reagents. We also thank Christine Portal and Quentin Lucas from INTS for their valued technical support in laboratory analysis. CTT, OD, DM, SL, and EM designed the study; CTT, DF, CA, ET, PNB, and SL conducted the study; CTT, CNT, and EM performed statistical analysis; CTT and GNT wrote the manuscript; and DM, SL, and EM revised the manuscript.

CONFLICT OF INTEREST

The authors have disclosed no conflicts of interest.

REFERENCES

1. Lefrère JJ, Dahourouh H, Dokekias AE, et al. Estimate of the residual risk of transfusion-transmitted human immunodeficiency virus infection in sub-Saharan Africa: a multinational collaborative study. Transfusion 2011;51:486-92.
2. Tagny CT, Kouao MD, Touré H, et al. Transfusion safety in Francophone African countries: an analysis of strategies for the medical selection of blood donors. Transfusion 2012;52:134-43.
3. Tagny CT, Diarra A, Yahaya R, et al. Characteristics of blood donors and donated blood in sub-Saharan Francophone Africa. Transfusion 2009;49:1592-9.
4. Tagny CT, Owusu-ofori S, Mbanya D, et al. The blood donor in sub-Saharan Africa: a review. Transfus Med 2010;20:1-10.
5. Schutz R, Savarit D, Kadjo JC, et al. Excluding blood donors at high risk of HIV infection in a West African city. BMJ 1993;307:1517-9.
6. McFarland W, Mvere D, Shamu R, et al. Risk factors for HIV seropositivity among first-time blood donors in Zimbabwe. Transfusion 1998;38:279-84.
7. Sentjens RE, Sisay Y, Vrielink H, et al. Prevalence of and risk factors for HIV infection in blood donors and various population subgroups in Ethiopia. Epidemiol Infect 2002;128:221-8.
8. Zachariah R, Harries AD, Nkhoma W, et al. HIV prevalence and demographic risk factors in blood donors. East Afr Med J 2002;79:88-91.
9. Cunha L, Plouzeau C, Ingrand P, et al. Use of replacement blood donors to study the epidemiology of major blood-borne viruses in the general population of Maputo, Mozambique. J Med Virol 2007;79:1832-40.

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10. Nebié KY, Olinger CM, Kafando E, et al. [Lack of knowledge among blood donors in Burkina Faso (West Africa); potential obstacle to transfusion security]. Transfus Clin Biol 2007;14: 446-52. French.
11. Tagny CT, Ndoumba A, Laperche S, et al. Reducing risks of transfusion-transmitted infections in a resource-limited hospital-based blood bank: the case of the Yaoundé University Teaching Hospital, Cameroon. ISBT Sci Series 2016;11:82-7.
12. Tante Yokyu E, Tagny CT, Mbanya D. Evaluation of risk factors of TTIs using blood donors' interviews. Cameroon: Faculty of Medicine and Biomedical Sciences, University of Yaoundé I; 2012.
13. Mole S, Onana E, Biholong D. [HIV and risk factors for the blood donors at the central hospital of Yaoundé, Cameroon]. Bull Soc Pathol Exot 2011;104:226-31. French.
14. Mbanya DN, Tagny CT. Blood safety begins with safe donations: update among blood donors in Yaoundé, Cameroon. Transfus Med 2005;5:395-9.
15. Tagny CT, Mbanya D, Leballais L, et al. Reduction of risk of transfusion-transmitted human immunodeficiency virus (HIV) infection by using an HIV antigen/antibody combination assay in blood donation screening in Cameroon. Transfusion 2011;51:184-90.
16. Minga A, Dohoun L, Abo Y, et al. Risk behaviors in volunteer blood donors who seroconverted for HIV, Abidjan, Côte d'Ivoire 1997 to 2005. Transfusion 2010;50: 888-93.
17. Noubiap JJ, Joko WY, Nansseu JR, et al. Sero-epidemiology of human immunodeficiency virus, hepatitis B and C viruses, and syphilis infections among first-time blood donors in Edéa, Cameroon. Int J Infect Dis 2013;17:832-7.
18. Mapako T, Mvere DA, Chitiyo ME, et al. Human immunodeficiency virus prevalence, incidence, and residual transmission risk in first-time and repeat blood donations in Zimbabwe: implications on blood safety. Transfusion 2013; 53:2413-21.
19. Sellors JW, Hayward R, Swanson G, et al. Comparison of deferral rates using a computerized versus written blood donor questionnaire: a randomized, cross-over study [ISRCTN84429599]. BMC Public Health 2002;2:14-6.
20. Custer B, Kessler D, Vahidnia F, et al. Risk factors for retrovirus and hepatitis virus infections in accepted blood donors. Transfusion 2015;55:1098-107.
21. Croce F, Fedeli P, Dahoma M, et al. Risk factors for HIV/AIDS in a low HIV prevalence site of sub-Saharan Africa. Trop Med Int Health 2007;12:1011-7.
22. Kabinda JM, Michèle DW, Donnen P, et al. Factors for viral infection in blood donors of South Kivu in the Democratic Republic of Congo. Pan Afr Med J 2014;19:385-8.
23. Matsuda EM, Coelho LP, Pimentel VF, et al. An HIV-1 transmission case possibly associated with manicure care. AIDS Res Hum Retroviruses 2014;30:1150-3.
24. Kucirka LM, Sarathy H, Govindan P, et al. Risk of window period HIV infection in high infectious risk donors: systematic review and meta-analysis. Am J Transplant 2011;11:1176-87.
25. de Almeida-Neto C, Goncalez TT, Birch RJ, et al. Risk factors for human immunodeficiency virus infection among Brazilian blood donors: a multicentre case-control study using audio computer-assisted structured interviews. Vox Sang 2013;105:91-9.
26. Musto JA, Seed CR, Law M, et al. Estimating the risk of blood donation associated with HIV risk behaviours. Transfus Med 2008;18:49-54.
27. Slot E, Janssen MP, Marijt-van der Kreek T, et al. Two decades of risk factors and transfusion-transmissible infections in Dutch blood donors. Transfusion 2016;56:203-14.
28. Kouegnigan RL, Biyoghe AS, Bengone C, et al. Evaluation of blood donor's questionnaire in a developing country: the case of Gabon. Transfus Clin Biol 2014;21:116-9.
29. Blatyta PF, Custer B, Gonçalez TT, et al. Undisclosed human immunodeficiency virus risk factors identified through a computer-based questionnaire program among blood donors in Brazil. Transfusion 2013;53:2734-43.
30. Katz LM, Cumming PD, Wallace EL. Computer-based blood donor screening: a status report. Transfus Med Rev 2007;21: 13-25.
31. Katz LM, Cumming PD, Wallace EL, et al. Audiovisual touchscreen computer-assisted self-interviewing for donor health histories: results from two years experience with the system. Transfusion 2005;45:171-80.

15372995, 2017, 8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/trf.14140 by EBMG ACCESS - GHANA, Wiley Online Library on [30/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License